Partial Derivatives

MATH 145 – Applied Calculus

Torin Quinlivan

October 2, 2024

Derivatives of Single Variable Functions

So far, everything we have discussed applies to functions of a single variable, of the form $f(x)$. What if we want to expand this to multivariable functions?

How do we extend our thoughts? For instance, what is the equivalent of a "tangent line" in three dimensions?

Derivatives are "rates of change", how would you define rates of change for a function in three dimensions?

Partial Derivatives of Functions $f(x, y)$

With functions $f(x, y)$ of *two variables*, you have the choice of *directions* (a.k.a. "traces")–which are parallel to the x-axis or the y-axis.

Each of these gives rise to a **partial derivative**:

$$\frac{\partial f}{\partial x}(a, b) = f_x(a, b) = \lim_{h \to 0} \frac{f(a+h, b) - f(a, b)}{h}$$
$$\frac{\partial f}{\partial y}(a, b) = f_y(a, b) = \lim_{h \to 0} \frac{f(a, b+h) - f(a, b)}{h}$$

Example: Find $\dfrac{\partial f}{\partial x}(2, 1)$ and $\dfrac{\partial f}{\partial y}(2, 1)$ if $f(x, y) = x^2 + y^2$.

Example: Find $\dfrac{\partial f}{\partial x}(2, 1)$ and $\dfrac{\partial f}{\partial y}(2, 1)$ if $f(x, y) = 3x + 5xy + y^2$.

Partial Derivatives of Functions $f(x, y)$

We can also take higher order partial derivatives:

$$\frac{\partial^2 f}{\partial x^2}(a, b) = f_{xx}(a, b) = \lim_{h \to 0} \frac{\dfrac{\partial f}{\partial x}(a+h, b) - \dfrac{\partial f}{\partial x}(a, b)}{h}$$

$$\frac{\partial^2 f}{\partial y^2}(a, b) = f_{yy}(a, b) = \lim_{h \to 0} \frac{\dfrac{\partial f}{\partial y}(a, b+h) - \dfrac{\partial f}{\partial y}(a, b)}{h}$$

$$\frac{\partial^2 f}{\partial x \partial y}(a, b) = f_{xy}(a, b) = \lim_{h \to 0} \frac{\dfrac{\partial f}{\partial x}(a, b+h) - \dfrac{\partial f}{\partial x}(a, b)}{h}$$

$$\frac{\partial^2 f}{\partial y \partial x}(a, b) = f_{yx}(a, b) = \lim_{h \to 0} \frac{\dfrac{\partial f}{\partial y}(a+h, b) - \dfrac{\partial f}{\partial x}(a, b)}{h}$$

Note: For most continuous functions, $\dfrac{\partial^2 f}{\partial x \partial y} = \dfrac{\partial^2 f}{\partial y \partial x}$.

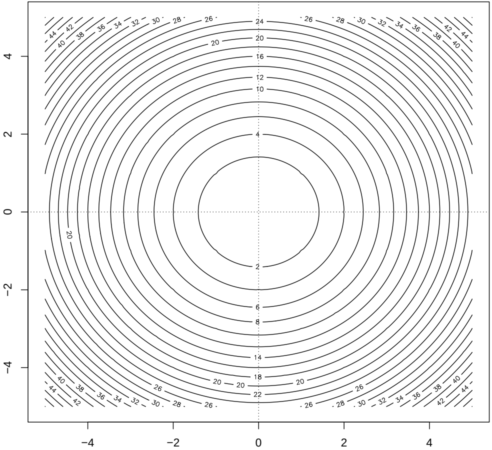
Partial Derivatives of Functions $f(x, y)$

You can also extend this concept to higher dimensional functions, $f(x_1, \ldots, x_n)$:

$$\frac{\partial f}{\partial x_i}(a_1, a_2, \ldots, a_{n-1}, a_n) = f_{x_i}(x_1, \ldots, x_n)$$
$$= \lim_{h \to 0} \frac{f(a_1, \ldots, a_i + h, \ldots, a_n) - f(a_1, \ldots, a_i, \ldots, a_n)}{h}$$

Partial Derivatives using Contour Plot

Example: Find $\dfrac{\partial f}{\partial x}(2,\,2)$ and $\dfrac{\partial f}{\partial y}(2,\,2)$.



Example: The *wind chill* is a function $w(v, T)$ of the wind speed v (in $^{m}/_{hr}$) and temperature T (in degrees F). Estimate $w_T(20, -15)$ and $w_v(20, -15)$. Explain what they mean.

$v\backslash T$	-30	-25	-20	-15	-10	-5	0	5	10	15	20
5	-46	-40	-34	-28	-22	-16	-11	-5	1	7	13
10	-53	-47	-41	-35	-28	-22	-16	-10	-4	3	9
15	-58	-51	-45	-39	-32	-26	-19	-13	-7	0	6
20	-61	-55	-48	-42	-35	-29	-22	-15	-9	-2	4
25	-64	-58	-51	-44	-37	-31	-24	-17	-11	-4	3
30	-67	-60	-53	-46	-39	-33	-26	-19	-12	-5	1
35	-69	-62	-55	-48	-41	-34	-27	-21	-14	-7	0
40	-71	-64	-57	-50	-43	-36	-29	-22	-15	-8	-1

Partial Derivatives and Concavity

- $\dfrac{\partial f}{\partial x} = f_x(a, b)$ represents the rate of change of $f(x, y)$ in the x-direction at the point (a, b).
- $\dfrac{\partial f}{\partial y} = f_y(a, b)$ represents the rate of change of $f(x, y)$ in the y-direction at the point (a, b).

- $\dfrac{\partial^2 f}{\partial x^2} = f_{xx}(a, b)$ represents the concavity of $f(x, y)$ in the x-direction at the point (a, b).
- $\dfrac{\partial^2 f}{\partial y^2} = f_{yy}(a, b)$ represents the concavity of $f(x, y)$ in the y-direction at the point (a, b).

Partial Derivative Functions

If we want the function form of the partial derivative, treat the variable not being derived on as a constant and derive as if it was a single variable function.

When computing ∂_x , hold y constant and use the usual rules for finding derivatives as a function of x.

When computing ∂_y , hold x constant and use the usual rules for finding derivatives as a function of y.

Example: Partial Derivatives

Example: Let $f(x,y) = yx^2 + 3xy^2$.

- Find $\dfrac{\partial f}{\partial x}$.
- Find f_y.

Example: Partial Derivatives

Example: Let $g(x,y) = \dfrac{xy + y^2 + x^2}{y + x}$.

- Find $\dfrac{\partial g}{\partial x}$.
- Find g_y.

Example: Higher-Order Partial Derivatives

Example: Find $\dfrac{\partial^2 f}{\partial y \partial x}(3,\, 2)$ if $f(x,\, y) = x^2 + y^2$.

Example: Higher-Order Partial Derivatives

Example: Let $h(x, y) = -4x^3y^2$.

- Find h_{xx}.
- Find h_{yy}.
- Find $\dfrac{\partial^2 h}{\partial x \partial y}$.
- Find $\dfrac{\partial^2 h}{\partial y \partial x}$.